End-TermExamination
(CBCS)(SUBJECTIVE TYPE)(OffLine)
Course Name: B. Tech. CSE-AI, Semester:<VI>
(April -May, 2023)

Subject Code: BAI 302	Subject: Natural Language Processing

Time :3 Hours Maximum Marks :60

Note:Q. 1 is compulsory. Attempt one question each from the Units I, II, III & IV.

Q1 (2.5*8=20)

(a) Explain Syntax Analysis (Parsing) with suitable example.
(b) Explain different phases of NLP with suitable examples of each phase.
(c) Assume the sentence below. Create Bag of Words (BOW) vectors before cleaning and after cleaning data/text.
S1: Welcome to great Learning, now start Learning
S2: Learning is a good practice
(d) Discuss Skip-gram model in details with suitable example.
(e) Comparison of Word-Based and Phrase-Based Machine Translation Systems
(f) What are the different types of chatbot architectures?
(g) Discuss the advantage and disadvantage of LSTM
(h) Explain the types of Sentiment Analysis andHow does Sentiment Analysis work?

UNIT-I

Q2 For the sentence "Hi my name is Michael. I am an enthusiastic Data Scientist. Currently I am working on a post about NLP, more specifically about the Pre-Processing Steps."Perform Sentence Tokenizer, Word Tokenizer and Stop Word Removal on this sentence. **(10)**

Q3 Explain Ambiguity , Challenges and explain its type with example **(10)**

UNIT-II

Q4 Assume the sentence below. Create Word2vec vectors using CBOW model and one–hot encode vector representation **(10)**
S1: I like NLP
S2: I like Deep Learning

Q5 Assume the sentence below. Create TF-IDF vectors using Term- document matrix **(10)**
S1: I do not like Vanila Cake
S2: : I do not like Vanilalcecream

UNIT-III

Q6 Explain the Working of conversational models, types of conversational models and future of conversational models. **(10)**

Q7 Discuss the various approaches of Machine Translation, Benefits of Machine Translation, its applications and Challenges in Machine Translation. **(10)**

UNIT-IV

Q8 Discuss Gated Recurrent Unit (GRU) architecture and also discuss its applications, Challenges and Limitations. **(10)**

Q9 Explain Types of RNN, its applications and discus advantage and disadvantage of RNN. **(10)**